

Rayarala Archana Sumanjali

Data Analyst

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SUMMARY

Data Analyst with 3 years of experience specializing in mortgage finance and cloud-based analytics. Adept at transforming raw data into actionable insights through advanced data wrangling, statistical analysis, and predictive modeling. Proficient in SQL for querying complex datasets, Python and R for data manipulation and automation, and Tableau/Power BI for building scalable, interactive dashboards. Experienced in designing ETL pipelines, performing exploratory data analysis (EDA), conducting A/B testing, and implementing time series forecasting models to drive strategic decision-making. Known for delivering high-impact visualizations and automating reporting workflows to improve operational efficiency across cross-functional teams.

SKILLS SET

Languages & Tools: Python, SQL (MySQL, SQL Server, Oracle), R, Excel (Pivot Tables, Macros)

Libraries: Pandas, NumPy, Scipy, Matplotlib, Seaborn, ggplot2

Visualization: Tableau, Power BI, SAS Visual Analytics

ETL Tools: SSIS, Talend, Apache NiFi, Hadoop

Databases: SQL Server, Oracle, MySQL, MongoDB

Version Control & Dev Tools: Git, GitHub, JIRA, Confluence, VS Code, PyCharm

CRM & Analytics Platforms: Salesforce, Google Analytics, SAP

Techniques: EDA, A/B Testing, Statistical Analysis, Time Series Forecasting, Predictive Modeling, Feature Engineering, Anomaly Detection, Data Storytelling

Methodologies: Agile, SDLC, Data Pipeline Automation

EXPERIENCE

Freddie Mac, USA | Data Analyst

02/2024 – Present

- Built advanced Tableau dashboards to monitor mortgage performance, identify high-risk loans, and visualize default trends, improving visibility for executive decision-makers.
- Developed predictive models using Python (scikit-learn, statsmodels) to assess early warning indicators for loan delinquency and prepayment behavior.
- Designed and automated ETL pipelines using SSIS and SQL Server to ingest financial data from multiple sources, enhancing data freshness and reliability.
- Processed large datasets using Pandas and NumPy for data wrangling, transformation, and feature engineering to support model input preparation.
- Leveraged SAS Visual Analytics to generate detailed risk and performance reports for internal audit and compliance teams.
- Integrated Salesforce CRM data into analytics pipelines to evaluate customer retention and loan origination trends, improving data segmentation and campaign strategy.

Hellinex Cloud, India | Data Analyst

09/2021 – 07/2022

- Conducted EDA using Python and R to analyze customer behavior patterns, product usage, and retention metrics across SaaS products.
- Created dynamic dashboards in Tableau and Power BI for stakeholders to monitor digital marketing KPIs, including CTR, conversion rate, and engagement.
- Built and maintained automated ETL workflows using Talend and Apache NiFi, improving reporting cadence and accuracy.
- Wrote and optimized complex SQL queries for marketing funnel analysis, improving dashboard load time by 20%.
- Led multivariate A/B tests across paid media campaigns to evaluate strategy effectiveness, resulting in a 30% increase in conversions.
- Used Google Analytics and Salesforce data to model user journeys, derive channel attribution insights, and recommend performance optimization strategies.
- Developed Excel-based tools and visualizations using formulas, pivot charts, and macros for presenting campaign results to leadership.

EDUCATION

M.S. in Data Science

Lewis University – Illinois, USA

Aug 2022 – Dec 2023

B.Tech in Computer Science

Jawaharlal Nehru Technological University – Hyderabad, India

Jul 2017 – Dec 2021